

Week 7- Cloud vulnerability Scanning

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- ❖ TASK- 1) Scan cloud instances using free scanners.
2) Identify vulnerabilities.
3) Document remediation steps.

Step 1: Log in to AWS Console

- Go to AWS Management Console.
- Sign in with your AWS account.

Step 2: Open AWS Inspector

- In the **search bar** (top of AWS console), type **Inspector**.
- Click on **Amazon Inspector**.

Step 3: Enable Inspector (First-time setup)

- If this is your **first time**, you'll see a **Get started** or **Enable Amazon Inspector** button.
- Click **Enable** → Wait for AWS to activate it

Step 4: Select the Instances for Scanning

- Amazon Inspector automatically detects your **EC2 instances** and **container images**.
- You'll see your resources listed under "**Coverage**".

Step 5: Run a Scan

- AWS Inspector scans automatically once enabled.
- If you want **manual scan**, click:
 - **Scans** → **Start scan** → Select resources → Click **Run**.

Step 6: View Vulnerability Findings

- Go to **Findings** tab.

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- You'll see a list of vulnerabilities:
 - **Severity:** Critical, High, Medium, Low.
 - **Description:** What the issue is.
 - **Affected Resource:** The EC2 instance ID.
 - **Remediation:** How to fix it.

Step 7: Document the Findings

In your documentation, note:

- **Finding name**
- **Severity**
- **Affected Resource**
- **Recommendation**

Step 8: Apply Remediation

Typical remediation steps:

- **Outdated packages** → SSH into the EC2 instance and run:

```
bash
CopyEdit
sudo apt update && sudo apt upgrade -y    # For Ubuntu/Debian
sudo yum update -y                        # For Amazon Linux/RHEL
```

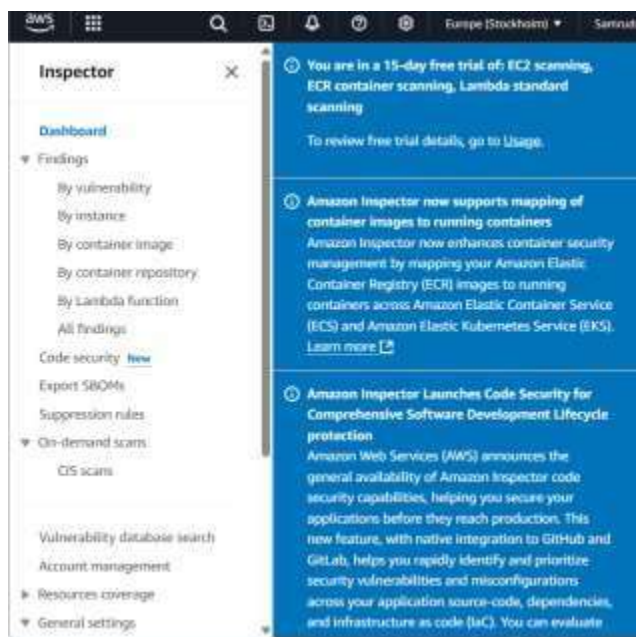
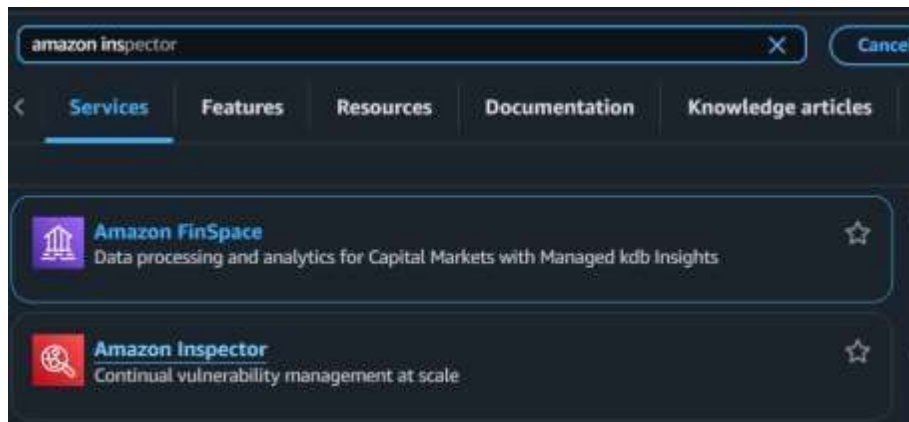
- **Unnecessary open ports** → Go to **EC2** → **Security Groups** → Edit inbound rules → Remove unused ports.
- **Weak configurations** → Update IAM policies, use stronger encryption, etc.

Step 9: Re-Scan After Fix

- Run another scan from **Inspector** → **Scans** → **Start Scan** to confirm vulnerabilities are resolved.

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❖ Screenshot-



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Findings: All findings [info](#)

All findings ranked by severity.

Findings (48) [Refresh](#) [Export findings](#) [Create suppression rule](#)

Choose a row to see the finding details.

Finding status: **Active** Filter criteria:

Severity	Title	Impacted resource	Type	Age	Status
High	CVE-2024-49861 - kernel, kernel-tool	i-0db0c8949e96fa372	Package Vulnerability	3 days	Active
High	CVE-2025-22869 - amazon-ssm-agent	i-0db0c8949e96fa372	Package Vulnerability	3 days	Active
High	CVE-2025-37890 - kernel, kernel-tool	i-0db0c8949e96fa372	Package Vulnerability	3 days	Active
High	CVE-2025-37942 - kernel, kernel-tool	i-0db0c8949e96fa372	Package Vulnerability	3 days	Active
High	CVE-2025-6021 - libxml2, libxml2-py	i-0db0c8949e96fa372	Package Vulnerability	3 days	Active
High	CVE-2025-38079 - kernel, kernel-tool	i-0db0c8949e96fa372	Package Vulnerability	3 days	Active
High	CVE-2025-38075 - kernel, kernel-tool	i-0db0c8949e96fa372	Package Vulnerability	3 days	Active

aws [Menu](#)

EC2 [Back](#)

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- Instances**
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
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 - Reserved Instances
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 - Capacity Reservations

```
Microsoft Windows [Version 10.0.22631.5699]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Shree>ssh -i ssh "C:\Users\Shree\Downloads\Mykeypair.pem"
Warning: Identity file ssh not accessible: No such file or directory.
ssh: Could not resolve hostname C:\\Users\\Shree\\Downloads\\Mykeypair.pem: No such host is known.

C:\Users\Shree>ssh -i "C:\Users\Shree\Downloads\Mykeypair.pem" ec2-user@ec2-16-170-242-131.eu-north-1.compute.amazonaws.com
The authenticity of host 'ec2-16-170-242-131.eu-north-1.compute.amazonaws.com (16.170.242.131)' can't be established.
ED25519 key fingerprint is SHA256:WIqa0pr+OdL1lAuqix/edcCcDM55sB+4sRFcmVSTZY4.
This host key is known by the following other names/addresses:
  C:\Users\Shree/.ssh/known_hosts:1: 16.170.242.131
Are you sure you want to continue connecting (yes/no/[fingerprint])? |
```

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```
C:\Users\Shree>ssh -i "C:\Users\Shree\Downloads\Mykeypair.pem" ec2-user@ec2-16-170-242-131.eu-north-1.compute.amazonaws.com
The authenticity of host 'ec2-16-170-242-131.eu-north-1.compute.amazonaws.com ([16.170.242.131])' can't be established.
ED25519 key fingerprint is: SHA256:W1qa0pr+OdL1Auqj/edcCdM55sB+4sRFcmVSTZY4.
This host key is known by the following other names/addresses:
  C:\Users\Shree\.ssh\known_hosts:1: 16.170.242.131
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-16-170-242-131.eu-north-1.compute.amazonaws.com' (ED25519) to the list of known hosts.
Last login: Thu Jun  5 07:44:15 2025 from 152.57.57.253
```

```
graph TD
    Root["#_"] --> Etc["####_"]
    Root --> Bin["#####_"]
    Etc --> Passwd["###|"]
    Bin --> Bash["#/"]
    Passwd --> V["V-"]
    Bash --> V
    V --> M["m/"]
```

Amazon Linux 2

AL2 End of Life is 2026-06-30.

A newer version of Amazon Linux is available!

Amazon Linux 2023, GA and supported until 2028-03-15.
<https://aws.amazon.com/linux/amazon-linux-2023/>

```
28 package(s) needed for security, out of 30 available
Run "sudo yum update" to apply all updates.
[ec2-user@ip-172-31-46-230 ~]$
```

```
verifying : openssl-server-1.4p1-22.amzn2.0.5.x86_64
```

```
Installed:
  kernel.x86_64 0:5.10.239-236.958.amzn2
```

Updated:

```
amazon-ssm-agent.x86_64 0:3.3.2299.0-1.amzn2
amazon-tools.x86_64 0:5.10.239-236.958.amzn2
libxml2.x86_64 0:2.9.1-6.amzn2.5.19
openssh.x86_64 0:7.4p1-22.amzn2.0.10
perl.x86_64 4:5.16.3-299.amzn2.0.3
perl-macros.x86_64 4:5.16.3-299.amzn2.0.3
python-lib.x86_64 0:2.7.18-1.amzn2.0.13
python2-cryptography.x86_64 0:1.7.2-2.amzn2.0.1
python3-lib.x86_64 0:3.7.16-1.amzn2.0.18
python.x86_64 0:4.1.0-0.27.20180314git3c2946.amzn2.0.2
```

```
aws-cfn-bootstrap.noarch 0:2.0-35.amzn2
libicu.x86_64 0:50.2.4.amzn2.0.2
libxml2-python.x86_64 0:2.9.1-6.amzn2.5.19
openssl-clients.x86_64 0:7.4p1-22.amzn2.0.10
perl-Pod-Escapes.noarch 1:1.04-299.amzn2.0.3
python.x86_64 0:2.7.18-1.amzn2.0.13
python-requests.noarch 0:2.6.8-10.amzn2.0.7
python2-setuptools.noarch 0:41.2.0-4.amzn2.0.6
python3-pip.noarch 0:20.2.2-1.amzn2.0.12
sudo.x86_64 0:1.8.23-10.amzn2.3.8
```

```
cloud-init.noarch 0:19.3-46.am
libtasn1.x86_64 0:4.10-1.amzn
ntr.x86_64 2:0.92-2.amzn2.0.2
openssh-server.x86_64 0:7.4p1-
perl-libs.x86_64 4:5.16.3-299.
python-devel.x86_64 0:2.7.18-1
python-urlgrab3.noarch 0:1.25.9
python3.x86_64 0:3.7.16-1.amzn
python3-setuptools.noarch 0:49
```

Complete!

```
[ec2-user@ip-172-31-46-230 ~]$
```

Activate Windows
Go to Settings to activate Windows.